OpenAI shipped code to production with **zero manually-written code**.

7 engineers · 5 months · **1M** AI-generated code · **10×** faster

Harness Engineering

The discipline that made it possible

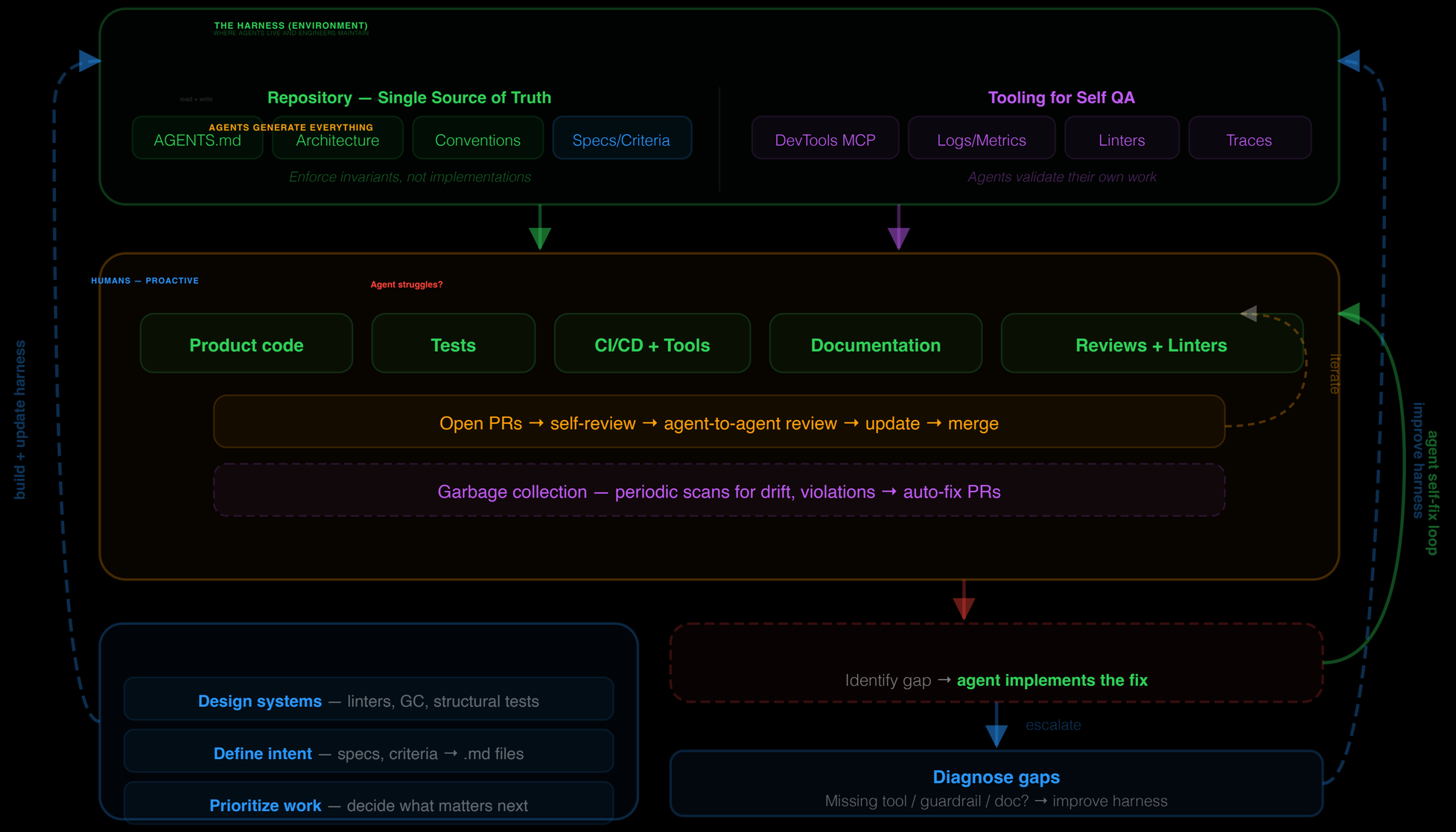
Yan Cheng · March 6, 2026

Every tech movie ever...

Scrolling logs. Progress bar. One person watching.

We're actually getting there.

Agents, humans, and the harness



Trial → error → better harness

✕ 1000-line AGENTS.md	→	✓ ~100-line map	· discover on demand
✕ Manual cleanup Fridays	→	✓ Automated GC	· agents scan + fix drift
✕ Context in Slack & Docs	→	✓ Everything in the repo	· single source of truth

"When the agent struggled, they didn't try harder — they encoded what was missing."

Transforming to harness engineering?

Changed skills and requirements from engineers

Environment Design
Structure repos + tooling for agent navigation

Prompt Decomposition
Break goals into agent-tractable units

Feedback Loops
Linters that teach fixes
Tests that enforce architecture

Gap Diagnosis
Agent fails? Find what's missing, then encode it

Transforming to harness engineering?

How to get there

For engineers

- Mindset shift — more architectural thinking
- Start small — cleanup, tests, docs
- Learning and sharing
- Debugging skills transfer directly

What we need from leadership

- Celebrate environment design wins
- Tooling time = product time
- Expect a slow start — it's investment
- Track how improvements reduce future failures

Let's be honest

Long-term coherence
5 months proven.
Years? Unknown.

Human leverage points
Where does our judgment add the most value?

Model capability growth
Today's constraints: unnecessary or critical?

Discipline hasn't disappeared — the hard work shifted from writing code to **designing the systems that make agents reliable**.

Where do we start?

Bring knowledge into the repo
It may live in Confluence, Teams, or someone's head

Make existing tools agent-readable
Monitoring, CI, linting — we have them, now make them accessible to agents

Pick a low-stakes pilot
Internal tooling, test generation, doc updates

Treat every agent failure as a system problem
Don't blame the model — fix the environment

Want AI work reliably for us?

But first, we need to work for AI —

through better context, better scaffolding, better environments.

That's **harness engineering**.

> *"Our most difficult challenges now center on designing environments,*
> *not expecting better models to solve everything."*
>
> — OpenAI Codex Team

My journey toward harness engineering

Building context engine for data-workspace
DnA — making our repo agent-navigable with structured context

PoC for session recording
Work-related — exploring agent-assisted capture and replay

Production web app — zero manual code
Hobby project: Danish exam prep app, fully AI-generated

danskprep.vercel.app